MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. If the angle made by the rope with the ground level is 30°, the height of the pole is: [1]

(A) 10 m (B) 20 m (C) $10\sqrt{3}$ m (D) 5 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $h = 20 \sin 30° = 20 \times \frac{1}{2} = \boxed{10 \text{ m}}$.

Q2. If the angles of elevation of the top of a tower from two points on the ground at distances a and b from the base are 30° and 60° respectively, then the height of the tower is: [1]

(A) $\sqrt{a+b}$ (B) $\sqrt{ab}$ (C) $\frac{a+b}{2}$ (D) ab

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: $\tan 30° = \frac{h}{a}$ and $\tan 60° = \frac{h}{b}$. Multiplying: $\tan 30° \tan 60° = \frac{h^2}{ab} \implies 1 = \frac{h^2}{ab} \implies h = \boxed{\sqrt{ab}}$.

Q3. A tree breaks due to a storm and the broken part bends so that the top of the tree touches the ground making an angle of 30° with it. The distance from the foot of the tree to the point where the top touches the ground is 8 m. The total height of the tree is: [1]

(A) $8\sqrt{3}$ m (B) $\frac{8\sqrt{3}}{3}$ m (C) $16\sqrt{3}$ m (D) 8 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Standing part $= 8 \tan 30° = \frac{8}{\sqrt{3}}$ m; broken slant part $= \frac{8}{\cos 30°} = \frac{16}{\sqrt{3}}$ m. Total height $= \frac{24}{\sqrt{3}} = \boxed{8\sqrt{3} \text{ m}}$.

Q4. The angle of depression of a point on the ground from the top of a 60 m high building is 45°. The distance of the point from the foot of the building is: [1]

(A) 30 m (B) 60 m (C) $60\sqrt{3}$ m (D) $30\sqrt{2}$ m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Alternate elevation is 45°, so $d = 60\cot 45° = \boxed{60 \text{ m}}$.

Q5. **Assertion (A):** A man standing on a platform 3 m above the surface of a lake observes a cloud and its reflection in the lake. The angle of elevation of the cloud is always strictly less than the angle of depression of its reflection.

Reason (R): The reflection is located at a depth below the water surface equal to the height of the cloud above the water surface, making the vertical distance from the observer to the reflection greater than that to the cloud. [1]

(A) Both (A) and (R) are true; (R) explains (A).

(B) Both are true; (R) does not explain (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: If cloud height above water is $H > 3$ m at horizontal distance x, elevation α satisfies $\tan\alpha = \dfrac{H-3}{x}$ whereas reflection depression β satisfies $\tan\beta = \dfrac{H+3}{x} > \tan\alpha \implies \alpha < \beta$. Both (A) and (R) are true and (R) explains (A).

SECTION B: Short Answer Questions

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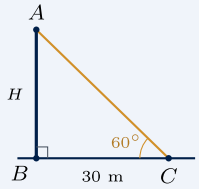
Q6. A 6 m high vertical pole casts a shadow $2\sqrt{3}$ m long on the ground. At the same time, a nearby tower casts a shadow 30 m long. Find the height of the tower. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

At the same instant, sun's elevation θ is equal for pole (6 m, shadow $2\sqrt{3}$ m) and tower (H m, shadow 30 m). [0.5 Mark]

For pole: $\tan\theta = \dfrac{6}{2\sqrt{3}} = \sqrt{3}$ ($\theta = 60°$). [1 Mark]

For tower: $\tan\theta = \dfrac{H}{30} = \sqrt{3} \implies H = 30\sqrt{3} = 30(1.732) = \boxed{30\sqrt{3} \text{ m} \approx 51.96 \text{ m}}$. [0.5 Mark]



Q7. From the top of a 60 m high building, the angles of depression of the top and bottom of a transmission tower are observed to be 30° and 60° respectively. Find the height of the transmission tower. **[3]**

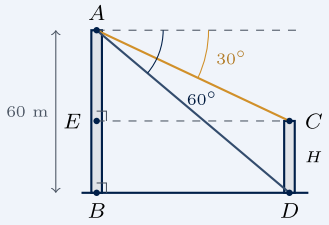
► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let building $AB = 60$ m and tower $CD = H$ m. Draw $CE \perp AB$ so $BE = CD = H$ and $AE = 60 - H$. By alternate angles, $\angle ACE = 30°$ and $\angle ADB = 60°$. [0.5 Mark]

In right $\triangle ABD$: $\tan 60° = \dfrac{60}{BD} \implies BD = CE = \dfrac{60}{\sqrt{3}} = 20\sqrt{3}$ m. [1 Mark]

In right $\triangle AEC$: $\tan 30° = \dfrac{AE}{CE} \implies AE = \dfrac{20\sqrt{3}}{\sqrt{3}} = 20$ m. [1 Mark]

Height of transmission tower $CD = BE = AB - AE = 60 - 20 = \boxed{40 \text{ m}}$. [0.5 Mark]



Q8. A man on top of a vertical tower 150 m high spots two cars parked on the same straight road leading to the tower. If their angles of depression are 30° and 60° respectively, find the distance between the two cars. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

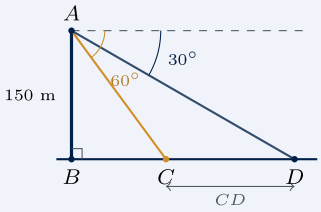
Let tower $AB = 150$ m and cars C, D lie on the same side along the road leading to foot B ($\angle ACB = 60°, \angle ADB = 30°$). [0.5 Mark]

In right $\triangle ABC$: $BC = \dfrac{150}{\tan 60°} = \dfrac{150}{\sqrt{3}} = 50\sqrt{3}$ m. [1 Mark]

In right $\triangle ABD$: $BD = \dfrac{150}{\tan 30°} = 150\sqrt{3}$ m. [1 Mark]

Distance between the two cars $= CD = BD - BC = 150\sqrt{3} - 50\sqrt{3} = 100\sqrt{3} = 100(1.732) = \boxed{100\sqrt{3} \text{ m} \approx 173.2 \text{ m}}$. [0.5 Mark]

Note: If the two cars are considered on opposite sides of the tower along the straight road, then $CD = 150\sqrt{3} + 50\sqrt{3} = 200\sqrt{3}$ m $\approx$ 346.4 m.



Q9. A boy of height 1.3 m spots a balloon moving with the wind in a horizontal level at some height from the ground. The angle of elevation of the balloon from his eyes at any instant is 60°. After 2 seconds, the angle of elevation reduces to 30°. If the horizontal speed of the wind is $29\sqrt{3}$ m/s, find the height of the balloon from the ground. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Horizontal distance travelled in 2 s is $d = 29\sqrt{3} \times 2 = 58\sqrt{3}$ m. Let h' be the balloon's vertical height above the boy's eye level (1.3 m). [0.5 Mark]

At 60°: $OB_{\text{eye}} = \dfrac{h'}{\sqrt{3}}$; at 30°: $OD_{\text{eye}} = h'\sqrt{3}$. [1 Mark]

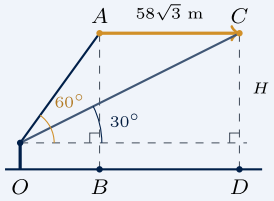
$$d = h'\sqrt{3} - \frac{h'}{\sqrt{3}} = \frac{2h'}{\sqrt{3}} = 58\sqrt{3}$$

Equating horizontal travel: [0.5 Mark]

$\implies 2h' = 58 \times 3 \implies h' = 87$ m. Adding boy's height (1.3 m):

Total height of balloon from ground $H = 87 + 1.3 = \boxed{88.3 \text{ m}}$

Final balloon height: [1 Mark]



Q10. CASE STUDY: Skysails Green Maritime Propulsion

A commercial cargo vessel deploys a kite sail A that ascends to an operating vertical height of 200 m above sea level ($AB = 200$ m). The tow cable is anchored to the vessel's bow at C, making an angle of inclination $\theta = 30°$ with the horizontal.

(i) Find the length of the tow cable paid out (AC) when the kite operates at an inclination of 30° to the horizontal. [1]

(ii) Find the horizontal distance between the vessel's bow (C) and point B directly beneath the kite. [1]

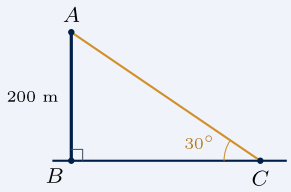
(iii) If changing wind patterns cause the cable inclination to increase to 45° while maintaining the same vertical height of 200 m, find the new length of the tow cable paid out. [Take $\sqrt{2} = 1.414$] [2]

OR

Find the decrease in horizontal distance between the vessel's bow and point B when the inclination increases from 30° to 45°. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) In right $\triangle ABC$ ($\angle ACB = 30°$, $AB = 200$ m):



$$\sin 30° = \frac{200}{AC} \implies \frac{1}{2} = \frac{200}{AC} \implies AC = \boxed{400 \text{ m}}$$

Subpart (i): [1 Mark]

(ii) In right $\triangle ABC$:

$$\tan 30° = \frac{200}{BC} \implies BC = 200\sqrt{3}$$

$$\implies BC = 200(1.732) = \boxed{200\sqrt{3} \text{ m} \approx 346.4 \text{ m}}$$

Subpart (ii): [1 Mark]

(iii) Main Choice: At inclination 45° and height 200 m, $\sin 45° = \frac{200}{AC'}$. [1 Mark]

$$\implies AC' = 200\sqrt{2} = 200 \times 1.414 = \boxed{200\sqrt{2} \text{ m} \approx 282.8 \text{ m}}$$

New cable length: [1 Mark]

(iii) OR Choice: At 45°, horizontal distance $BC' = 200 \cot 45° = 200$ m. [1 Mark]

$$\text{Decrease} = 200\sqrt{3} - 200 = 200(1.732 - 1) = \boxed{146.4 \text{ m}}$$

Decrease in horizontal distance: [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★